

FLYN AI Enterprise CRM Blueprint

1. System Overview

FLYN AI is an AI-powered conversational operating system integrating WhatsApp, Telegram, AI agents, CRM, automation, and APIs.

Architecture is multi-tenant, event-driven, and AI-native.

2. Core Modules

- Messaging Engine (WhatsApp + Telegram)
- AI Engine (LLM-based agents)
- Automation Engine (Flows + Webhooks)
- CRM/CDP Layer
- Campaign Engine
- Developer API Layer
- SaaS Admin Layer

3. Messaging Architecture

Unified inbox system with cross-channel conversation merging.

Real-time sync, message status tracking, and internal collaboration.

Supports agent routing, tagging, and conversation history.

4. AI Engine

AI Receptionist, Sales Agent, and Support Agent.

Intent detection, sentiment analysis, and multilingual capabilities.

Context-aware responses using CRM data.

5. Automation & Webhooks

Event-driven system with triggers from internal and external sources.

Supports Shopify, WooCommerce, and custom APIs.
Includes retry logic, logging, and secure signing.

6. CRM & Data Model

Centralized customer profiles with lifecycle tracking.
Lead scoring, segmentation, and custom fields.
Unified identity across all channels.

7. Campaign System

Broadcast messaging with segmentation.
A/B testing and AI optimization.
Performance tracking and ROI analytics.

8. API & Developer Platform

REST APIs, Webhooks, and SDK tokens.
Swagger documentation and developer portal.
Rate limiting and monitoring.

9. Multi-Tenant SaaS Design

Organization-based isolation using `organization_id`.
Supports white-labeling, billing, and usage tracking.

10. Database Schema (High-Level)

Tables:

- organizations
- users
- contacts
- conversations
- messages

- campaigns
- workflows
- webhook_logs
- ai_interactions

11. Implementation Phases

Phase 1: Messaging + CRM Core

Phase 2: AI Engine Integration

Phase 3: Automation & Webhooks

Phase 4: Campaigns & Analytics

Phase 5: API & Developer Platform

Phase 6: Scaling & Optimization

12. Positioning

FLYN AI competes with Intercom, HubSpot, Twilio, and GoHighLevel.

It unifies communication, automation, AI, and CRM into one platform.